GATE DA 2027

Programming, Data Structures & Algorithms

Topic-wise Question Bank

Weighted by what GATE DA 2024–2026 actually asked

232 practice questions + **9** real previous-year questions • **10** topics

MCQ • MSQ • NAT — the three GATE question types

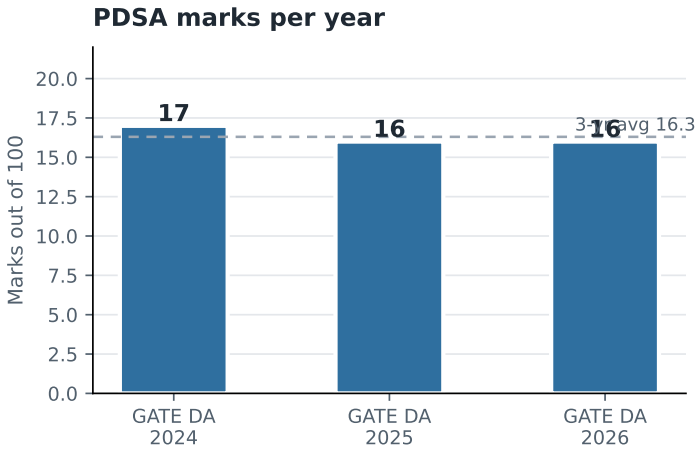
Answer key at the end — answers only, no worked solutions

Every code snippet in this book was **executed in Python** and its printed output compared with the stored answer before the key was fixed — all 47 executions agreed. The 9 previous-year questions carry the official GATE answer key. Practice options are shuffled, so the key is not all "A".

Prepared for Ashutosh • August 2026

How this book is built

PDSA is the single most stable, highest-weight subject in GATE DA. Reading the official question papers directly, Programming & DSA carried **17 marks in 2024**, **16 marks in 2025** and **16 marks in 2026** — a three-year average of **16.3 marks**, and the most predictable band of any subject (it never moves more than a mark). The planning range for 2027 is a tight **15 to 18 marks**. If you are triaging what to master first, it is this.

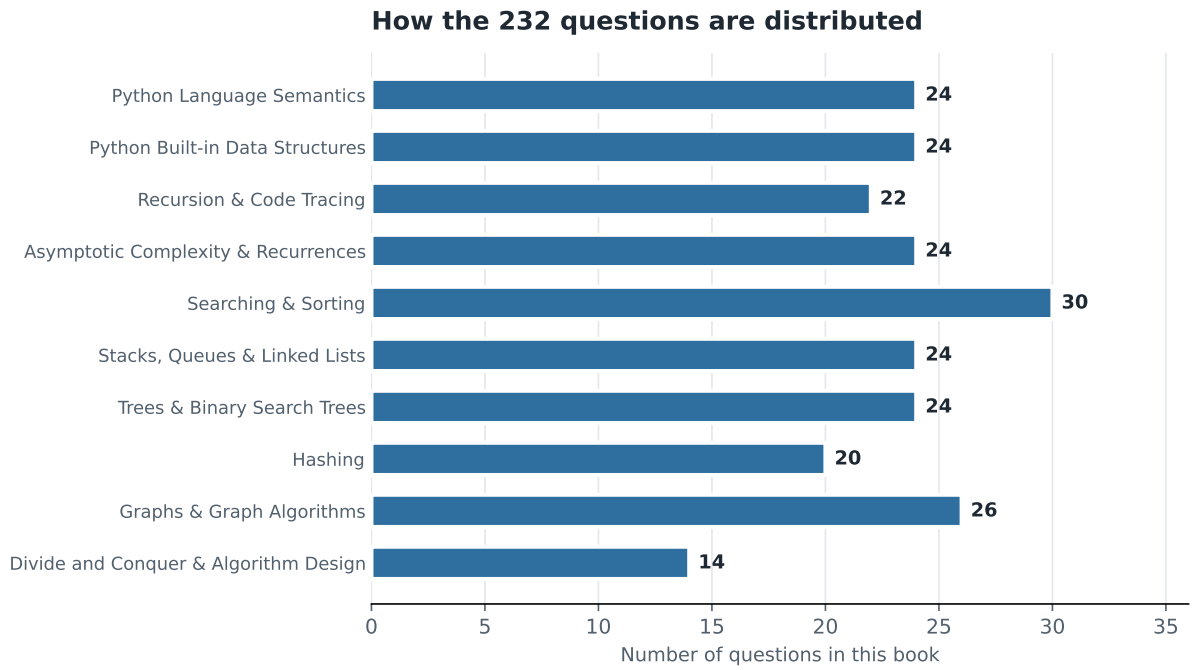


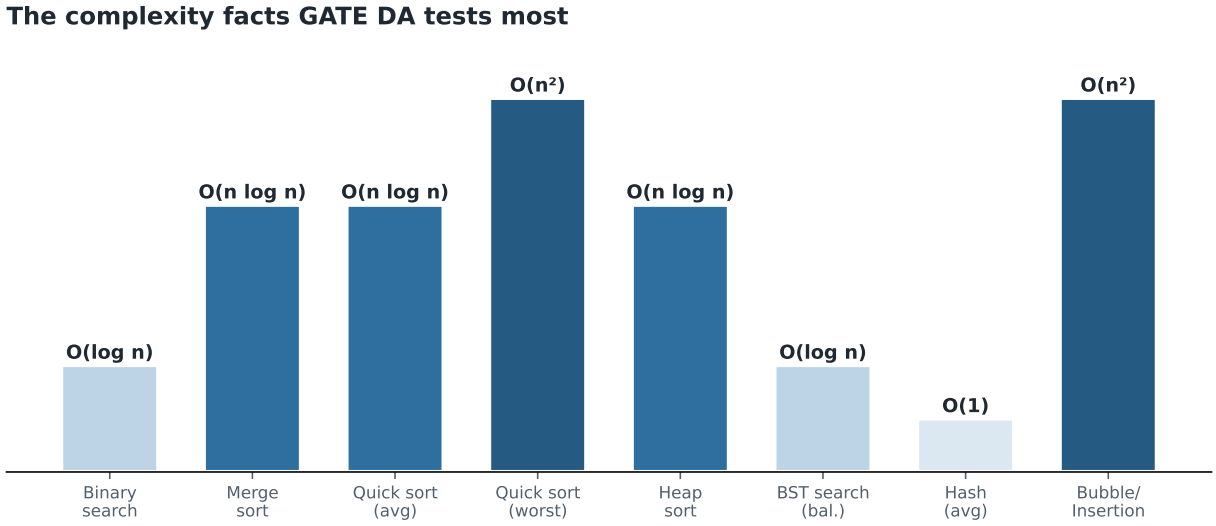
What GATE DA actually asks — and it is not theory

This is the part most people get wrong. GATE DA's PDSA is *not* the definitions-and-theorems DSA of a classical CS paper. Looking at every PDSA question in the official 2024, 2025 and 2026 papers, four habits stand out:

1. **"What does this print?" dominates.** You are handed a short *Python* program and asked for its exact output, a count, or the number of times something happens. 2026 alone had the mutable-default-argument trap, a closure sharing a list, a recursive stack-activation count, and a recursive bubble-pass count. This is code tracing, not recall.
2. **It is Python-specific.** Mutable default arguments, aliasing, closures capturing a variable, list slicing and in-place mutation — these are tested *because* they are Python. Know cold what `def f(x, lst=[])` does on the second call.
3. **Complexity is asked as mechanics, not proofs.** "How many comparisons does binary search make in the worst case on 1000 elements?", "which recurrence matches this quick-sort?", "how many swaps after two passes of bubble sort?". You count, you don't derive.
4. **Data-structure questions are about behaviour.** Given operations on a stack, a BST, a hash table with a stated collision policy — what is the final state? The entries never matter; the procedure does.

The number of questions given to each topic below follows that reality: Python semantics, recursion/tracing, complexity, and searching & sorting carry the most weight, because that is where the marks have actually been.



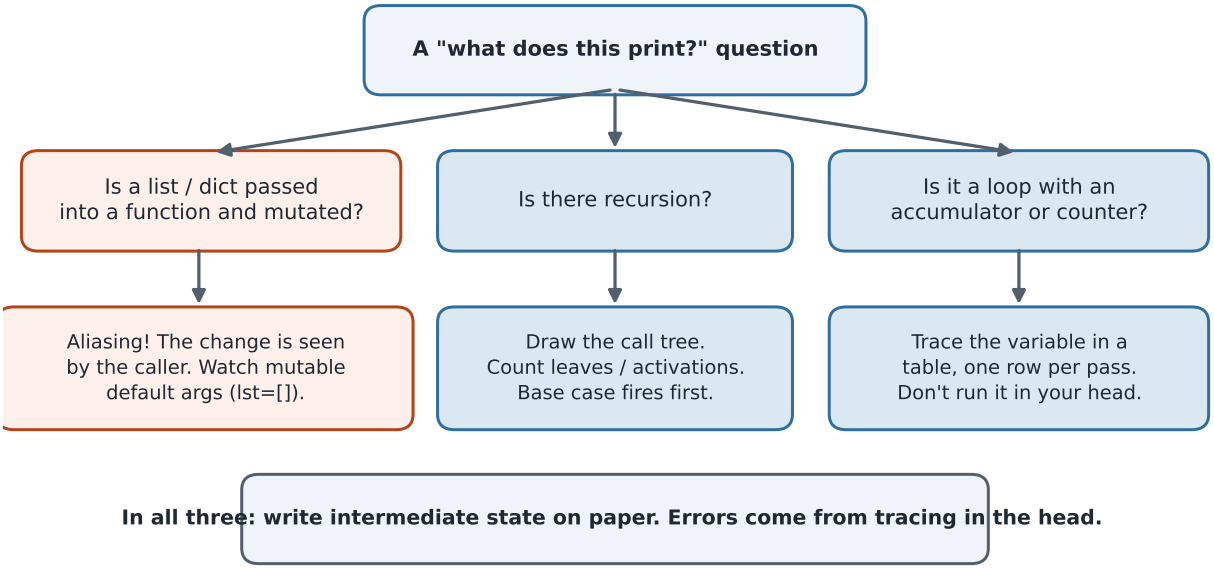


Question types and marking (same as the real paper)

Type	Description	Negative marking
MCQ	four options, exactly one correct	$-1/3$ (1-mark) or $-2/3$ (2-mark)
MSQ	four options, one *or more* correct; no partial credit	none
NAT	numerical answer, entered on a virtual keypad	none

This bank is 34% MCQ, 13% MSQ and 53% NAT. The extra NAT weighting is deliberate: NAT carries no negative marking, and "how many swaps / comparisons / calls" is exactly the counting drill GATE DA rewards.

How to attack a "what does this print?" question



Twelve facts that answer a disproportionate number of PDSA questions.

- A mutable default argument (`def f(x, lst=[])`) is created *once* and *persists* across calls — the classic trap.
- Lists, dicts and sets are passed by reference: mutating inside a function is visible outside. Rebinding (`lst = [...]`) is not.
- A closure captures the *variable*, not its value at definition time.
- Binary search on n elements makes at most $\lfloor \log_2 n \rfloor + 1$ comparisons; for $n = 1000$ that is 10.
- Merge sort is $\Theta(n \log n)$ always; quicksort is $\Theta(n \log n)$ average but $\Theta(n^2)$ worst (sorted input with first/last pivot).
- Master theorem: $T(n) = aT(n/b) + \Theta(n^d)$ compares d with $\log_b a$.
- A stack is LIFO, a queue is FIFO; a recursive call uses the runtime *stack*.
- In a BST, an inorder traversal returns the keys in sorted order.
- A balanced BST / heap has height $\Theta(\log n)$; a skewed BST degrades to $\Theta(n)$.
- Hashing with chaining: average $O(1)$, worst $O(n)$ when all keys collide.
- Number of leaf calls of the naive Fibonacci recursion for $F(n)$ is $F(n{+}1)$.
- In-place `a, b = b, a` swaps simultaneously — the right side is evaluated first.

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Part A — Practice questions by topic

1. Python Language Semantics

24 questions • 8 MCQ, 3 MSQ, 13 NAT • Suggested time: 55 minutes

Q.1

[NAT • 1 Mark • Easy]

Consider the Python code below.

```
a = [1, 2, 3]
b = a
b.append(4)
print(len(a))
```

The value printed is _____.

Q.2

[MCQ • 1 Mark • Easy]

Which one of the following Python objects is **immutable**?

(A) tuple (B) dict (C) set (D) list

Q.3

[NAT • 1 Mark • Easy]

Consider the Python code below.

```
x = [1, 2]
y = x[:]
y.append(3)
print(len(x))
```

The value printed is _____.

Q.4

[MCQ • 1 Mark • Easy]

In Python, the expression `type([]) is list` evaluates to

(A) True (B) a TypeError (C) None (D) False

Q.5

[NAT • 2 Marks • Moderate]

Consider the Python code below.

```
def f(x, acc=[]):
acc.append(x)
return acc

f(1)
print(len(f(2)))
```

The value printed is _____.

Q.6 [MCQ • 1 Mark • Moderate]

Consider the Python code below.

```
x = 10
def g():
x = 20
g()
print(x)
```

The value printed is

(A) 20 (B) nothing — it raises an error

(C) None (D) 10

Q.7 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
c = 5
def h():
global c
c += 1
h()
h()
print(c)
```

The value printed is _____.

Q.8 [MSQ • 1 Mark • Moderate]

Which of the following Python built-in types is/are **mutable**?

(A) set (B) dict (C) tuple (D) list

Q.9 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
import copy
a = [[1, 2], [3, 4]]
b = copy.copy(a)
b[0].append(9)
print(len(a[0]))
```

The value printed is _____.

Q.10 [MCQ • 2 Marks • Moderate]

Consider the Python code below.

```
a = [1, 2, 3]
b = a
a = a + [4]
print(len(b))
```

The value printed is

(A) it raises an error (B) 0 (C) 4 (D) 3

Q.11 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
a = [1, 2, 3]
b = a
a += [4]
print(len(b))
```

The value printed is _____.

Q.12 [MCQ • 1 Mark • Moderate]

For `a = [1, 2]` and `b = [1, 2]`, the expression `(a == b, a is b)` evaluates to

(A) (False, False) (B) (True, True) (C) (False, True) (D) (True, False)

Q.13 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
vals = [0, 1, '', None, 'a', [], [0]]
print(sum(1 for v in vals if v))
```

The value printed is _____.

Q.14 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about Python functions is/are correct?

(A) A default argument expression is evaluated once, when the function is defined

(B) Rebinding a parameter name inside a function changes the caller's variable

(C) Arguments are passed by object reference

(D) A function can return several values packed as a tuple

Q.15 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
def f(n):
n = n + [1]

x = [0]
f(x)
print(len(x))
```

The value printed is _____.

Q.16 [MCQ • 1 Mark • Moderate]

For `s = äbc¨`, the statement `s[0] = 'z'` results in

(A) a TypeError, because strings are immutable (B) `s` becoming `żbc¨`

(C) `s` becoming `z̈` (D) a SyntaxError

Q.17 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
fs = [lambda: i for i in range(3)]
print(fs[0]())
```

The value printed is _____.

Q.18 [NAT • 1 Mark • Moderate]

Consider the Python code below.

```
print(len([x for x in range(10) if x % 3 == 0]))
```

The value printed is _____.

Q.19 [MCQ • 1 Mark • Moderate]

In Python, the expression `(x for x in range(3))` produces

(A) a list (B) a set (C) a generator object (D) a tuple

Q.20 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
print(len([(i, j) for i in range(3) for j in range(i)]))
```

The value printed is _____.

Q.21 [MCQ • 1 Mark • Moderate]

In Python 3, the expression `(7 / 2, 7 // 2)` evaluates to

(A) (3, 3.5) (B) (3, 3) (C) (3.5, 3.5) (D) (3.5, 3)

Q.22 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
a = [[0] * 3] * 2
a[0][0] = 5
print(sum(a[1]))
```

The value printed is _____.

Q.23 [MSQ • 2 Marks • Moderate]

Given a list `a`, which of the following expression(s) create(s) a **new** list object rather than another name for the same object?

(A) `a[:]` (B) `list(a)` (C) `a` (D) `a.copy()`

Q.24 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
def f(a, b=[]):
b.append(a)
return b

f(1)
f(2)
print(len(f(3)))
```

The value printed is _____.

2. Python Built-in Data Structures

24 questions • 8 MCQ, 2 MSQ, 14 NAT • Suggested time: 54 minutes

Q.1 [NAT • 1 Mark • Easy]

The value of `len({1, 2, 2, 3, 3, 3})` is _____.

Q.2 [MCQ • 1 Mark • Easy]

Which list method removes and returns the **last** element of a list?

(A) `del()` (B) `pop()` (C) `remove()` (D) `discard()`

Q.3 [NAT • 1 Mark • Easy]

Consider the Python code below.

```
print(len("hello world".split()))
```

The value printed is _____.

Q.4 [MCQ • 1 Mark • Easy]

Which one of the following cannot be used as a key in a Python dictionary?

(A) a list (B) a tuple of integers (C) a string (D) an integer

Q.5 [NAT • 1 Mark • Moderate]

Consider the Python code below.

```
A = [1, 2, 3]
B = [4, 5, 6]
A.extend(B)
print(len(A))
```

The value printed is _____.

Q.6 [MCQ • 2 Marks • Moderate]

Given `A = [1, 2, 3]` and `B = [4, 5, 6]`, which single operation makes `A` equal to `[1, 2, 3, 4, 5, 6]`?

(A) `A.extend(B)` (B) `A.update(B)`

(C) `A.insert(3, B)` (D) `A.append(B)`

Q.7 [NAT • 1 Mark • Moderate]

Consider the Python code below.

```
d = {'a': 1, 'b': 2}
d['c'] = 3
print(len(d))
```

The value printed is _____.

Q.8 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about Python sets is/are correct?

(A) They are unordered

(B) They cannot contain duplicate elements

(C) They support indexing, e.g. `s[0]`

(D) Their elements must be hashable

Q.9 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
A = {1, 2, 3, 4}
B = {3, 4, 5}
print(len(A - B) + len(A & B))
```

The value printed is _____.

Q.10 [MCQ • 1 Mark • Moderate]

The expression `sorted([3, 1, 2]) == [1, 2, 3]` evaluates to

(A) True (B) None (C) a TypeError (D) False

Q.11 [NAT • 1 Mark • Moderate]

Consider the Python code below.

```
s = "abracadabra"
print(s.count("a"))
```

The value printed is _____.

Q.12 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
print(len(set("mississippi")))
```

The value printed is _____.

Q.13 [MCQ • 1 Mark • Moderate]

Python tuples support

(A) item assignment, but not slicing

(B) neither indexing nor slicing

(C) item assignment and appending

(D) indexing and slicing, but not item assignment

Q.14 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
a = list(range(10))
print(len(a[2:8:2]))
```

The value printed is _____.

Q.15 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
d = {x: x * x for x in range(5) if x % 2}
print(sum(d.values()))
```

The value printed is _____.

Q.16 [MCQ • 1 Mark • Easy]

For `a = [1, 2, 3]`, the value of `a[-1]` is

(A) an IndexError (B) 3 (C) 1 (D) 2

Q.17 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
A = {1, 2, 3}
B = {2, 3, 4}
print(len(A ^ B))
```

The value printed is _____.

Q.18 [MSQ • 2 Marks • Moderate]

Which of the following can be used as keys in a Python dictionary?

(A) `'x'` (B) `(1, 2)` (C) `3.5` (D) `[1, 2]`

Q.19 [NAT • 1 Mark • Moderate]

Consider the Python code below.

```
a = [5, 3, 1]
a.sort()
print(a[0])
```

The value printed is _____.

Q.20 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
d = {}
for ch in "banana":
d[ch] = d.get(ch, 0) + 1
print(d['a'])
```

The value printed is _____.

Q.21 [MCQ • 1 Mark • Moderate]

The value of the Python expression `äbc[::-1]` is

(A) `'cba'` (B) `'bca'` (C) `'abc'` (D) a SyntaxError

Q.22 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
a = [1, 2, 3, 4, 5]
print(sum(a[i] for i in range(len(a)) if i % 2 == 0))
```

The value printed is _____.

Q.23 [MCQ • 2 Marks • Moderate]

In Python 3.9 and later, the expression `d1 | d2` for two dictionaries produces

(A) a TypeError

(B) a new dictionary merging both, with `d1` winning on shared keys

(C) a new dictionary merging both, with `d2` winning on shared keys

(D) the set intersection of the keys

Q.24 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
s = set()
for i in range(10):
s.add(i % 4)
print(len(s))
```

The value printed is _____.

3. Recursion & Code Tracing

22 questions • 6 MCQ, 2 MSQ, 14 NAT • Suggested time: 54 minutes

Q.1

[NAT • 1 Mark • Easy]

Consider the Python code below.

```
def fact(n):
if n <= 1:
return 1
return n * fact(n - 1)

print(fact(5))
```

The value printed is _____.

Q.2

[MCQ • 1 Mark • Easy]

The purpose of a base case in a recursive function is to

(A) make the function run faster

(B) stop the recursion and return without a further recursive call

(C) allow the function to be called from other functions

(D) reduce the memory used by each call

Q.3

[NAT • 1 Mark • Moderate]

Consider the Python code below.

```
def fib(n):
if n < 2:
return n
return fib(n - 1) + fib(n - 2)

print(fib(6))
```

The value printed is _____.

Q.4

[NAT • 2 Marks • Moderate]

For the function `fib` defined in the previous question, the total number of invocations of `fib` (including the initial call) while evaluating `fib(5)` is _____.

Q.5

[NAT • 1 Mark • Moderate]

Consider the Python code below.

```
def gcd(a, b):
if b == 0:
return a
return gcd(b, a % b)

print(gcd(48, 18))
```

The value printed is _____.

Q.6 [MCQ • 1 Mark • Moderate]

A recursive function calls itself once with argument $n-1$ until n reaches 0. The maximum depth of the call stack is

(A) $\Theta(n^2)$ (B) $\Theta(1)$ (C) $\Theta(\log n)$ (D) $\Theta(n)$

Q.7 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
def f(n):
if n == 0:
return 0
return n % 10 + f(n // 10)

print(f(9875))
```

The value printed is _____.

Q.8 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
def f(a, b):
if a <= 0 or b <= 0:
return 0
if a > b:
return 1 + f(a - b, b)
return 1 + f(a, b - a)

print(f(15, 10))
```

The value printed is _____.

Q.9 [NAT • 2 Marks • Moderate]

The minimum number of moves needed to solve the Tower of Hanoi puzzle with 5 discs is _____.

Q.10 [MCQ • 1 Mark • Moderate]

Compared with an equivalent iterative version, a recursive implementation generally uses

(A) exactly the same stack space (B) more stack space

(C) no stack space at all (D) less stack space

Q.11 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
def f(n):
if n == 0:
return 0
return n % 10 + f(n // 10)

print(f(12345))
```

The value printed is _____.

Q.12 [MCQ • 2 Marks • Moderate]

Consider the Python code below.

```
def f(s):
if len(s) <= 1:
return s
return f(s[1:]) + s[0]

print(f("abcd"))
```

The string printed is

(A) `dcba` (B) `bcda` (C) `adcb` (D) `abcd`

Q.13 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about recursion is/are correct?

(A) Deep recursion can exhaust the call stack

(B) A terminating recursive function must have at least one base case

(C) A recursive solution is always faster than an iterative one

(D) Recursive calls consume space on the call stack

Q.14 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
def f(n):
if n <= 1:
return 1
return f(n - 1) + f(n - 2) + 1

print(f(5))
```

The value printed is _____.

Q.15 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
def ones(n):
if n == 0:
return 0
return n % 2 + ones(n // 2)

print(ones(29))
```

The value printed is _____.

Q.16 [MCQ • 2 Marks • Hard]

Consider the Python code below.

```
def rev(a, i, j):
if i >= j:
return
a[i], a[j] = a[j], a[i]
rev(a, i + 1, j - 1)
```

This function

(A) reverses the elements of `a` between indices `i` and `j` in place

(B) rotates `a` left by one position

(C) sorts the elements of `a` between indices `i` and `j`

(D) leaves `a` unchanged

Q.17 [NAT • 1 Mark • Moderate]

Consider the Python code below.

```
def power(b, e):
if e == 0:
return 1
return b * power(b, e - 1)

print(power(2, 10))
```

The value printed is _____.

Q.18 [NAT • 2 Marks • Hard]

For the standard recursion `fib(n) = fib(n-1) + fib(n-2)` with `fib(0)=0` and `fib(1)=1`, the number of times `fib(1)` is invoked while evaluating `fib(6)` is _____.

Q.19 [MCQ • 1 Mark • Moderate]

The depth of the recursion tree when merge sort is applied to an array of n elements is

(A) $\Theta(n)$ (B) $\Theta(\log n)$ (C) $\Theta(1)$ (D) $\Theta(n \log n)$

Q.20 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
def f(n, acc=0):
if n == 0:
return acc
return f(n - 1, acc + n)

print(f(10))
```

The value printed is _____.

Q.21 [MSQ • 2 Marks • Moderate]

Which of the following algorithms is/are naturally expressed by divide and conquer?

(A) Bubble sort (B) Merge sort (C) Quick sort (D) Binary search

---

Q.22 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
def f(s):
if not s:
return 0
return (1 if s[0] in 'aeiou' else 0) + f(s[1:])

print(f("programming"))
```

The value printed is _____.

---

4. Asymptotic Complexity & Recurrences

24 questions • 10 MCQ, 3 MSQ, 11 NAT • Suggested time: 61 minutes

Q.1 [MCQ • 1 Mark • Easy]

The worst-case time complexity of binary search on a sorted array of n elements is

(A) $\Theta(\log n)$ (B) $\Theta(1)$ (C) $\Theta(n \log n)$ (D) $\Theta(n)$

Q.2 [NAT • 1 Mark • Easy]

Let $T(n) = T(n/2) + 1$ with $T(1) = 0$, defined for n a power of 2. Then $T(64) =$ _____.

Q.3 [MCQ • 1 Mark • Easy]

The worst-case time complexity of linear search on an array of n elements is

(A) $\Theta(1)$ (B) $\Theta(n)$ (C) $\Theta(n^2)$ (D) $\Theta(\log n)$

Q.4 [NAT • 2 Marks • Moderate]

Let $T(n) = 2T(n/2) + n$ with $T(1) = 1$, defined for n a power of 2. Then $T(8) =$ _____.

Q.5 [MCQ • 1 Mark • Moderate]

The worst-case time complexity of merge sort is

(A) $\Theta(n \log n)$ (B) $\Theta(\log n)$ (C) $\Theta(n^2)$ (D) $\Theta(n)$

Q.6 [MCQ • 1 Mark • Moderate]

The worst-case time complexity of quick sort is

(A) $\Theta(\log n)$ (B) $\Theta(n)$ (C) $\Theta(n \log n)$ (D) $\Theta(n^2)$

Q.7 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
x = 0
for i in range(1, 6):
for j in range(i, 6):
x += 1
print(x)
```

The value printed is _____.

Q.8 [MSQ • 2 Marks • Moderate]

Which of the following sorting algorithms has/have $\Theta(n \log n)$ **worst-case** running time?

(A) Heap sort

(B) Insertion sort

(C) Merge sort

(D) Quick sort with the first element as pivot

Q.9 [NAT • 2 Marks • Moderate]

Let $T(n) = T(n-1) + n$ with $T(0) = 0$. Then $T(10) =$ _____.

Q.10 [MCQ • 2 Marks • Moderate]

By the master theorem, $T(n) = 4T(n/2) + n$ is

(A) $\Theta(n^2 \log n)$ (B) $\Theta(n^2)$

(C) $\Theta(n \log n)$ (D) $\Theta(n)$

Q.11 [NAT • 2 Marks • Hard]

Let $T(n) = 3T(n/3) + n$ with $T(1) = 1$, defined for n a power of 3. Then $T(27) =$ _____.

Q.12 [MCQ • 1 Mark • Moderate]

Which of the following functions grows fastest asymptotically?

(A) $n!$ (B) $n \log n$ (C) 2^n (D) n^2

Q.13 [NAT • 2 Marks • Moderate]

Consider the Python code below, with `n = 1000`.

```
n = 1000
c = 0
i = 1
while i < n:
c += 1
i *= 2
print(c)
```

The number of times the loop body executes is _____.

Q.14 [MSQ • 2 Marks • Hard]

Which of the following statement(s) about asymptotic notation is/are correct?

(A) O expresses an upper bound (B) Θ expresses a tight bound

(C) $n = O(n^2)$ (D) $n^2 = O(n)$

Q.15 [NAT • 2 Marks • Moderate]

The maximum number of comparisons made by binary search on a sorted array of 1000 elements is _____.

Q.16 [MCQ • 2 Marks • Moderate]

The **amortized** time of appending one element to a dynamic array that doubles its capacity when full is

(A) $O(n)$ (B) $O(1)$ (C) $O(\log n)$ (D) $O(n \log n)$

Q.17 [NAT • 2 Marks • Hard]

Consider the Python code below, with `n = 8`.

```
n = 8
c = 0
for i in range(n):
j = 1
while j < n:
c += 1
j *= 2
print(c)
```

The value printed is _____.

Q.18 [MCQ • 2 Marks • Moderate]

The auxiliary space complexity of the standard merge sort on an array of n elements is

(A) $\Theta(1)$ (B) $\Theta(n)$ (C) $\Theta(n \log n)$ (D) $\Theta(\log n)$

Q.19 [NAT • 2 Marks • Hard]

Let $T(n) = 2T(n-1) + 1$ with $T(0) = 0$. Then $T(5) =$ _____.

Q.20 [MCQ • 2 Marks • Moderate]

Selection sort on an array of n elements makes

(A) $\Theta(n)$ comparisons in the worst case

(B) $\Theta(n \log n)$ comparisons on average

(C) $\Theta(n)$ comparisons when the array is already sorted

(D) $\Theta(n^2)$ comparisons regardless of the input

Q.21 [NAT • 2 Marks • Hard]

The value of $\sum_{i=1}^{10} i^2$ is _____.

Q.22 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) is/are correct?

(A) $O(2n)$ and $O(n)$ denote the same growth class

(B) Constant factors are ignored in asymptotic notation

(C) $O(1)$ grows no faster than $O(\log n)$

(D) $O(n^2)$ and $O(n)$ denote the same growth class

Q.23 [NAT • 2 Marks • Hard]

Consider the Python code below, with `n = 6`.

```
n = 6
c = 0
for i in range(1, n + 1):
for j in range(1, n + 1, i):
c += 1
print(c)
```

The value printed is _____.

Q.24 [MCQ • 1 Mark • Easy]

Finding the maximum element of an unsorted array of n elements requires

(A) $\Theta(1)$ time

(B) $\Theta(n \log n)$ time

(C) $\Theta(n)$ time

(D) $\Theta(\log n)$ time

---

5. Searching & Sorting

30 questions • 11 MCQ, 4 MSQ, 15 NAT • Suggested time: 79 minutes

Q.1 [MCQ • 1 Mark • Easy]

Binary search can be applied to an array only if the array is

(A) of even length

(B) stored as a linked list

(C) sorted

(D) free of duplicates

Q.2 [NAT • 1 Mark • Easy]

The maximum number of comparisons made by binary search on a sorted array of 15 elements is _____.

Q.3 [MCQ • 1 Mark • Easy]

Bubble sort with an early-exit flag runs in which time on an already sorted array?

(A) $\Theta(n^2)$ (B) $\Theta(\log n)$ (C) $\Theta(n \log n)$ (D) $\Theta(n)$

Q.4 [NAT • 2 Marks • Moderate]

The number of swaps performed by insertion sort while sorting the array $[5, 4, 3, 2, 1]$ into increasing order is _____.

Q.5 [MSQ • 2 Marks • Moderate]

Which of the following sorting algorithms is/are **stable** in their standard implementation?

(A) Merge sort (B) Insertion sort (C) Bubble sort (D) Selection sort

Q.6 [NAT • 2 Marks • Moderate]

The number of comparisons made by selection sort on an array of 6 elements is _____.

Q.7 [MCQ • 2 Marks • Moderate]

The worst case of quick sort occurs when the chosen pivot is, at every step,

(A) chosen uniformly at random

(B) the median of the current subarray

(C) the smallest or the largest element of the current subarray

(D) the middle element by position

Q.8 [NAT • 2 Marks • Moderate]

The number of levels of recursion in merge sort applied to an array of 8 elements, counting the top-level call as level 0, is _____.

Q.9 [NAT • 2 Marks • Moderate]

The number of swaps performed by bubble sort while sorting $[4, 3, 2, 1, 5]$ into increasing order is _____.

Q.10 [MCQ • 1 Mark • Moderate]

Which of the following sorting algorithms is **not** in-place?

(A) Insertion sort (B) Selection sort (C) Merge sort (D) Quick sort

---

Q.11 [NAT • 2 Marks • Hard]

Quick sort with the *last* element as pivot is run on the already sorted array $[60, 70, 80, 90, 100]$. The total number of key comparisons performed is _____.

Q.12 [MSQ • 2 Marks • Moderate]

Which of the following sorting algorithms has/have $\Theta(n^2)$ worst-case running time?

(A) Insertion sort (B) Merge sort (C) Bubble sort (D) Selection sort

---

Q.13 [NAT • 1 Mark • Easy]

The maximum number of comparisons made by linear search on an array of 100 elements is _____.

Q.14 [MCQ • 2 Marks • Moderate]

Binary search on a sorted singly linked list of n nodes does **not** run in $O(\log n)$ time because

(A) the list may contain duplicates

(B) a linked list does not support constant-time access to the middle element

(C) comparisons on a linked list are slower

(D) a linked list cannot be sorted

Q.15 [NAT • 2 Marks • Hard]

Binary search is used to look for the value 37 in the sorted array $[1, 2, 3, \ldots, 100]$, with the middle index computed as $\lfloor (\text{lo} + \text{hi})/2 \rfloor$. The number of key comparisons for equality performed is _____.

Q.16 [MCQ • 1 Mark • Moderate]

Merging two sorted arrays of sizes m and n into one sorted array takes

(A) $\Theta(mn)$ time (B) $\Theta(\log(m + n))$ time

(C) $\Theta(m + n)$ time (D) $\Theta(m \log n)$ time

Q.17 [NAT • 2 Marks • Moderate]

The number of inversions in the array $[2, 4, 1, 3, 5]$ is _____.

Q.18 [NAT • 2 Marks • Hard]

Insertion sort is used on an array whose sorted prefix is $[2, 4, 6, 8]$, and the next element to be inserted is an integer x. The number of integer values of x in the range 1 to 20 for which exactly two elements must be shifted is _____.

Q.19 [MCQ • 1 Mark • Moderate]

The time complexity of heap sort is

(A) $\Theta(n^2)$ in the worst case

(B) $\Theta(\log n)$ in the average case

(C) $\Theta(n)$ in the best case

(D) $\Theta(n \log n)$ in the best, average and worst cases

Q.20 [NAT • 2 Marks • Hard]

The number of comparisons needed to find both the minimum and the maximum of an array of 100 elements, using the pairwise method, is _____.

Q.21 [MSQ • 2 Marks • Hard]

Which of the following statement(s) about quick sort is/are correct?

(A) Its worst-case running time is $\Theta(n^2)$

(B) Its average-case running time is $\Theta(n \log n)$

(C) It sorts in place

(D) It is a stable sorting algorithm

Q.22 [NAT • 2 Marks • Moderate]

Two sorted subarrays of sizes 4 and 3 are merged. The maximum number of comparisons the merge can require is _____.

Q.23 [MCQ • 2 Marks • Moderate]

The lower bound $\Omega(n \log n)$ on sorting applies to

(A) comparison-based sorting algorithms only

(B) every possible sorting algorithm

(C) only stable sorting algorithms

(D) only in-place sorting algorithms

Q.24 [NAT • 2 Marks • Hard]

Bubble sort is applied to $[5, 1, 4, 2, 8]$. The total number of swaps performed during the **first two passes** is _____.

Q.25 [MCQ • 2 Marks • Moderate]

Which sorting algorithm is generally the best choice for an array that is already *nearly* sorted?

(A) Heap sort

(B) Selection sort

(C) Quick sort with the first element as pivot (D) Insertion sort

---

Q.26 [NAT • 2 Marks • Hard]

The minimum number of comparisons needed to sort three distinct elements in the worst case is _____.

Q.27 [MCQ • 2 Marks • Moderate]

The number of comparisons made by selection sort on an array of n elements is

(A) $\frac{n(n-1)}{2}$ regardless of the input (B) $n-1$ in the best case

(C) n^2 exactly (D) $n \log n$ on average

Q.28 [MSQ • 2 Marks • Moderate]

Which of the following require(s) the input to be sorted in order to work correctly?

(A) Interpolation search

(B) The merge step of merge sort, applied to its two input subarrays

(C) Linear search

(D) Binary search

Q.29 [NAT • 2 Marks • Hard]

The number of swaps performed by selection sort on an array of 6 elements is at most _____.

Q.30 [MCQ • 2 Marks • Hard]

Counting sort achieves $\Theta(n+k)$ time on n integers in the range 0 to k because it

(A) does not compare keys with one another (B) sorts the array in place

(C) uses a balanced binary search tree (D) is a divide-and-conquer algorithm

---

6. Stacks, Queues & Linked Lists

24 questions • 8 MCQ, 3 MSQ, 13 NAT • Suggested time: 58 minutes

Q.1 [MCQ • 1 Mark • Easy]

A stack follows which discipline?

(A) Random access

(B) First In, First Out

(C) Last In, First Out

(D) Highest priority first

Q.2 [MCQ • 1 Mark • Easy]

A queue follows which discipline?

(A) Last In, First Out

(B) Random access

(C) First In, First Out

(D) Highest priority first

Q.3 [NAT • 1 Mark • Easy]

Values 1, 2 and 3 are pushed onto an empty stack in that order, and then two `pop` operations are performed. The value now at the top of the stack is _____.

Q.4 [NAT • 2 Marks • Moderate]

Consider the Python code below.

```
s = []
total = 0
for x in [3, 1, 4, 1, 5]:
if s and s[-1] > x:
total += s.pop()
s.append(x)
print(total)
```

The value printed is _____.

Q.5 [MCQ • 1 Mark • Moderate]

Which data structure is most appropriate for checking whether the parentheses in an expression are balanced?

(A) Queue (B) Binary search tree (C) Hash table (D) Stack

Q.6 [NAT • 2 Marks • Moderate]

In a circular queue implemented on an array of size 5, the front index is 2 and the rear index is 4, with the rear pointing at the last stored element. The number of elements currently in the queue is _____.

Q.7 [MSQ • 2 Marks • Moderate]

For which of the following is a **stack** the natural underlying data structure?

(A) Implementing an undo facility

(B) Evaluating a postfix expression

(C) Managing function calls and returns

(D) Breadth-first traversal of a graph

Q.8 [MCQ • 2 Marks • Moderate]

When a queue is implemented using two stacks, the **amortized** cost of a dequeue operation is

(A) $O(1)$ (B) $O(\log n)$ (C) $O(n^2)$ (D) $O(n)$

Q.9 [NAT • 1 Mark • Moderate]

The value of the postfix expression `5 3 + 2 *` is _____.

Q.10 [MCQ • 2 Marks • Moderate]

The postfix form of the infix expression $a + b \times c$ is

(A) `ab+c*` (B) `abc*+` (C) `a+bc*` (D) `abc+*`

Q.11 [NAT • 2 Marks • Moderate]

In a singly linked list, the number of pointer assignments required to insert a new node at the **head** is _____.

Q.12 [MCQ • 2 Marks • Moderate]

Given only a pointer to a node (which is not the last node) in a singly linked list, that node can be deleted in $O(1)$ time by

(A) setting its own pointer to `None`

(B) it cannot be done at all

(C) traversing the list from the head to find its predecessor

(D) copying the next node's data into it and unlinking the next node

---

Q.13 [NAT • 1 Mark • Moderate]

A doubly linked list has n nodes. The total number of pointer fields across all nodes is _____ (in terms of n, enter the coefficient of n).

Q.14 [NAT • 2 Marks • Hard]

The slow-and-fast pointer technique is used to find the middle node of a singly linked list with 15 nodes. The number of times the slow pointer advances is _____.

Q.15 [MSQ • 2 Marks • Moderate]

Comparing a singly linked list with a dynamic array, which of the following statement(s) is/are correct?

(A) A linked list generally has better cache locality

(B) An array occupies a contiguous block of memory

(C) Insertion at the head is $O(1)$ for the linked list

(D) Access by index is $O(1)$ for the array

Q.16 [NAT • 1 Mark • Moderate]

An array-based stack has capacity 4. The values $1, 2, 3, 4, 5$ are pushed in order, with a push on a full stack failing. The number of successful pushes is _____.

Q.17 [NAT • 2 Marks • Hard]

Consider the Python code below.

```
from collections import deque
q = deque()
for i in range(1, 6):
q.append(i)
for _ in range(2):
q.append(q.popleft())
print(q[0])
```

The value printed is _____.

Q.18 [MCQ • 2 Marks • Moderate]

An LRU cache with $O(1)$ lookup and $O(1)$ eviction is usually implemented using

(A) a queue alone

(B) a single sorted array

(C) a hash table together with a doubly linked list

(D) a binary search tree alone

Q.19 [NAT • 2 Marks • Hard]

A singly linked list of 10 nodes is reversed iteratively by redirecting each node's `next` pointer. The number of pointer redirections performed is _____.

Q.20 [MCQ • 1 Mark • Moderate]

A priority queue is most commonly implemented using

(A) a stack

(B) a binary heap

(C) a queue

(D) a singly linked list kept unsorted

Q.21 [NAT • 2 Marks • Hard]

Two stacks are implemented in a single array of size 100, growing towards each other from opposite ends. The maximum total number of elements they can hold together is _____.

Q.22 [NAT • 1 Mark • Moderate]

The value of the postfix expression `6 2 / 3 -` is _____.

Q.23 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about a circular queue of capacity n is/are correct?

(A) It avoids the wasted space of a simple linear queue

(B) Enqueue takes $O(n)$ time

(C) It is empty when front = rear

(D) Using one slot as a sentinel, it is full when $(\text{rear} + 1) \bmod n = \text{front}$

Q.24 [NAT • 2 Marks • Hard]

A stack is used to reverse a string of 12 characters by pushing every character and then popping every character. The total number of push and pop operations performed is _____.

---

7. Trees & Binary Search Trees

24 questions • 7 MCQ, 3 MSQ, 14 NAT • Suggested time: 61 minutes

Q.1 [MCQ • 1 Mark • Easy]

An inorder traversal of a binary search tree visits the keys in

(A) level order (B) increasing order (C) decreasing order (D) insertion order

---

Q.2 [NAT • 1 Mark • Easy]

The number of nodes in a perfect binary tree of height 3 (the root being at height 0) is _____.

Q.3 [MCQ • 1 Mark • Easy]

The root of a max-heap always contains

(A) the most recently inserted element (B) the minimum element

(C) the maximum element (D) the median element

Q.4 [NAT • 2 Marks • Moderate]

The keys $50, 30, 70, 20, 40, 60, 80$ are inserted, in that order, into an initially empty binary search tree. The number of leaf nodes in the resulting tree is _____.

Q.5 [NAT • 2 Marks • Moderate]

For the binary search tree of the previous question, the height, measured in edges on the longest root-to-leaf path, is _____.

Q.6 [MCQ • 2 Marks • Moderate]

For the same binary search tree built from $50, 30, 70, 20, 40, 60, 80$, the preorder traversal is

(A) $50, 30, 70, 20, 40, 60, 80$ (B) $20, 40, 30, 60, 80, 70, 50$

(C) $20, 30, 40, 50, 60, 70, 80$ (D) $50, 30, 20, 40, 70, 60, 80$

Q.7 [MSQ • 2 Marks • Moderate]

Which pair(s) of traversals is/are sufficient to reconstruct an arbitrary binary tree uniquely?

(A) Inorder alone (B) Inorder and preorder

(C) Inorder and postorder (D) Preorder and postorder

Q.8 [NAT • 2 Marks • Moderate]

A complete binary tree has 15 nodes. Its height, measured in edges, is _____.

Q.9 [MCQ • 1 Mark • Moderate]

The worst-case time to search for a key in a binary search tree with n nodes is

(A) $O(n \log n)$ (B) $O(1)$ (C) $O(n)$ (D) $O(\log n)$

Q.10 [NAT • 2 Marks • Moderate]

The number of structurally distinct binary search trees that can be built on 4 distinct keys is _____.

Q.11 [NAT • 2 Marks • Hard]

The number of structurally distinct binary trees with 5 nodes is _____.

Q.12 [MCQ • 2 Marks • Moderate]

The height of an AVL tree with n nodes is

(A) $O(\log n)$ (B) $O(\sqrt{n})$ (C) $O(n)$ (D) $O(1)$

Q.13 [NAT • 2 Marks • Moderate]

In a binary tree in which every internal node has exactly two children, there are 10 leaves. The number of internal nodes is _____.

Q.14 [NAT • 2 Marks • Hard]

A binary tree has inorder traversal $4, 2, 5, 1, 6, 3$ and preorder traversal $1, 2, 4, 5, 3, 6$. The right child of the root is the node with key _____.

Q.15 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about a binary min-heap of n elements is/are correct?

(A) It is a complete binary tree

(B) Extract-min takes $O(\log n)$ time

(C) It can be built from an unsorted array in $O(n)$ time

(D) Its inorder traversal lists the elements in increasing order

Q.16 [NAT • 1 Mark • Moderate]

The maximum possible number of nodes at level 4 of a binary tree, with the root at level 0, is _____.

Q.17 [MCQ • 1 Mark • Easy]

In a postorder traversal of a binary tree, the root is visited

(A) in level order

(B) first

(C) last

(D) between the two subtrees

Q.18 [NAT • 1 Mark • Moderate]

A tree with 20 nodes has _____ edges.

Q.19 [NAT • 2 Marks • Moderate]

The minimum possible height, measured in edges, of a binary tree with 100 nodes is _____.

Q.20 [MCQ • 2 Marks • Hard]

In a binary search tree, the inorder successor of a node that has a right subtree is

(A) the leftmost node of its right subtree

(B) the rightmost node of its right subtree

(C) the leftmost node of its left subtree

(D) its parent

Q.21 [NAT • 2 Marks • Moderate]

The keys $1, 2, 3, \ldots, 7$ are inserted in increasing order into an initially empty binary search tree. The height of the resulting tree, in edges, is _____.

Q.22 [NAT • 2 Marks • Hard]

In the binary search tree built from $50, 30, 70, 20, 40, 60, 80$, the number of key comparisons made while searching for 80 is _____.

Q.23 [MSQ • 2 Marks • Hard]

Which of the following statement(s) about binary trees is/are correct?

(A) Every binary tree is a binary search tree

(B) A tree with n nodes has exactly $n - 1$ edges

(C) A binary tree of height h has at most $2^{h+1} - 1$ nodes

(D) The maximum number of nodes at depth d is 2^d

Q.24 [NAT • 2 Marks • Hard]

In a level-order (breadth-first) traversal of the binary search tree built from $50, 30, 70, 20, 40, 60, 80$, the fifth node visited has key _____.

8. Hashing

20 questions • 7 MCQ, 3 MSQ, 10 NAT • Suggested time: 49 minutes

Q.1 [MCQ • 1 Mark • Easy]

A good hash function should

(A) map all keys to the same slot

(B) preserve the ordering of the keys

(C) be as slow as possible to compute

(D) distribute keys as uniformly as possible over the slots

Q.2 [NAT • 1 Mark • Easy]

With the hash function $h(k) = k \bmod 7$, the key 10 maps to slot _____.

Q.3 [MCQ • 1 Mark • Easy]

A collision in a hash table occurs when

(A) the hash function returns a negative value

(B) two distinct keys map to the same slot

(C) the table becomes full

(D) a key is deleted twice

Q.4 [NAT • 2 Marks • Moderate]

A hash table with 7 slots uses $h(k) = k \bmod 7$ with linear probing. The keys $10, 22, 31, 4, 15$ are inserted in that order. The key 4 finally occupies slot _____.

Q.5 [MCQ • 1 Mark • Moderate]

In separate chaining, collisions are resolved by

(A) storing all keys hashing to a slot in a list attached to that slot

(B) rehashing the entire table on every collision

(C) discarding the colliding key

(D) probing subsequent slots in the table

Q.6 [NAT • 1 Mark • Moderate]

A hash table with 20 slots currently stores 15 keys. Its load factor is _____ (rounded off to two decimal places).

Q.7 [MSQ • 2 Marks • Moderate]

Which of the following is/are open addressing schemes?

(A) Quadratic probing (B) Linear probing (C) Separate chaining (D) Double hashing

Q.8 [NAT • 2 Marks • Hard]

A hash table with 10 slots uses $h(x) = 3x \bmod 10$ with linear probing. The keys $1, 4, 5, 6, 14, 15$ are inserted in that order. The key 14 finally occupies slot _____.

Q.9 [MCQ • 2 Marks • Moderate]

The worst-case time to search for a key in a hash table with separate chaining and n keys is

(A) $O(\log n)$ (B) $O(n)$ (C) $O(1)$ (D) $O(n \log n)$

Q.10 [NAT • 2 Marks • Moderate]

Under the uniform hashing assumption with open addressing and load factor $\alpha = 0.5$, the expected number of probes for an **unsuccessful** search is _____.

Q.11 [NAT • 2 Marks • Hard]

A hash table with 10 slots uses $h(k) = k \bmod 10$ with quadratic probing, the i-th probe being at $(h(k) + i^2) \bmod 10$. The keys $10, 20, 30$ are inserted in that order. The key 30 finally occupies slot _____.

Q.12 [MCQ • 2 Marks • Moderate]

Primary clustering, in which long runs of occupied slots build up, is a characteristic drawback of

(A) separate chaining (B) perfect hashing (C) double hashing (D) linear probing

---

Q.13 [NAT • 2 Marks • Moderate]

A hash table with 11 slots uses $h(k) = k \bmod 11$ and separate chaining. The keys $22, 33, 44$ are inserted. The length of the chain at slot 0 is _____.

Q.14 [MSQ • 2 Marks • Hard]

Which of the following statement(s) about hash tables is/are correct?

(A) The keys can be listed in sorted order in $O(n)$ time

(B) Lookup takes $O(1)$ time on average under reasonable assumptions

(C) Keys must be hashable

(D) Lookup can take $O(n)$ time in the worst case

Q.15 [NAT • 2 Marks • Moderate]

A hash table with 7 slots uses double hashing with $h_1(k) = k \bmod 7$ and $h_2(k) = 1 + (k \bmod 5)$. The key 15 hashes to slot 1, which is already occupied. The next slot probed is _____.

Q.16 [MCQ • 1 Mark • Moderate]

Rehashing a hash table into a larger table is normally triggered when

(A) the hash function changes

(B) a key is deleted

(C) the first collision occurs

(D) the load factor exceeds a chosen threshold

Q.17 [NAT • 2 Marks • Hard]

A hash table with 10 slots uses separate chaining and stores 30 keys. The average chain length is _____.

Q.18 [MCQ • 2 Marks • Moderate]

A hash table is a poor choice when the application must frequently

(A) delete a single key

(B) retrieve all keys lying within a given range

(C) test membership of a single key

(D) insert a single key

Q.19 [NAT • 2 Marks • Hard]

A hash table with 5 slots uses $h(k) = k \bmod 5$ with linear probing. The keys $5, 10, 15$ are inserted in that order. The key 15 finally occupies slot _____.

Q.20 [MSQ • 1 Mark • Moderate]

Which of the following Python objects can be used as dictionary keys?

(A) a string

(B) a list of integers

(C) an integer

(D) a tuple of integers

---

9. Graphs & Graph Algorithms

26 questions • 9 MCQ, 4 MSQ, 13 NAT • Suggested time: 60 minutes

Q.1 [MCQ • 1 Mark • Easy]

The space required by the adjacency matrix representation of a graph with V vertices is

(A) $\Theta(V)$ (B) $\Theta(V+E)$ (C) $\Theta(V^2)$ (D) $\Theta(E)$

Q.2 [NAT • 1 Mark • Easy]

The number of edges in a complete undirected graph on 8 vertices is _____.

Q.3 [MCQ • 1 Mark • Easy]

Breadth-first search uses which auxiliary data structure?

(A) a priority queue (B) a stack (C) a queue (D) a hash table

Q.4 [MCQ • 1 Mark • Moderate]

Depth-first search is naturally implemented using

(A) a queue

(B) a priority queue

(C) a sorted array

(D) a stack, or equivalently recursion

Q.5 [NAT • 1 Mark • Easy]

A tree with 12 vertices has _____ edges.

Q.6 [NAT • 2 Marks • Moderate]

An undirected weighted graph has vertices $\{A, B, C, D, E\}$ and edges with weights $AB = 2$, $AC = 5$, $BC = 1$, $BD = 4$, $CD = 2$, $CE = 6$, $DE = 3$. The length of the shortest path from A to E is _____.

Q.7 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about breadth-first search is/are correct?

(A) It finds shortest paths in an unweighted graph

(B) It runs in $O(V+E)$ time with an adjacency list

(C) It uses a queue

(D) It finds shortest paths in a graph with arbitrary positive edge weights

Q.8 [NAT • 1 Mark • Easy]

A graph has 10 vertices and no edges. The number of connected components is _____.

Q.9 [MCQ • 1 Mark • Moderate]

Dijkstra's algorithm may produce incorrect results when the graph contains

(A) negative edge weights

(B) cycles

(C) isolated vertices

(D) more than one shortest path

Q.10 [NAT • 1 Mark • Moderate]

A minimum spanning tree of a connected graph with 9 vertices has _____ edges.

Q.11 [NAT • 2 Marks • Hard]

Consider the directed graph with vertex set $\{0, 1, \ldots, 20\}$ in which there is an edge from i to j exactly when $0 < j - i \leq 2$. A depth-first search starts at vertex 0 and always explores the smallest-numbered unvisited neighbour first. The number of vertices discovered *after* vertex 10 is _____.

Q.12 [MCQ • 1 Mark • Moderate]

A topological ordering of a directed graph exists if and only if the graph

(A) is acyclic

(B) is connected

(C) is complete

(D) has no isolated vertices

Q.13 [NAT • 2 Marks • Moderate]

A directed acyclic graph has vertices $\{A, B, C, D\}$ and edges $A \to B$, $A \to C$, $B \to D$, $C \to D$. The number of distinct topological orderings is _____.

Q.14 [MSQ • 2 Marks • Hard]

Which of the following statement(s) about Dijkstra's algorithm is/are correct?

(A) It computes shortest paths from a single source

(B) It handles negative edge weights correctly

(C) It is a greedy algorithm

(D) With a binary heap it runs in $O((V + E) \log V)$ time

Q.15 [NAT • 1 Mark • Moderate]

In a simple undirected graph the degrees of all vertices sum to 20. The number of edges is _____.

Q.16 [MCQ • 1 Mark • Moderate]

For a sparse graph, the adjacency list representation is preferred over the adjacency matrix mainly because

(A) it stores edge weights more accurately

(B) it makes the graph acyclic

(C) it uses $\Theta(V + E)$ space rather than $\Theta(V^2)$

(D) it allows edge lookup in $O(1)$ time

Q.17 [NAT • 2 Marks • Hard]

An undirected graph has edges $0-1$, $0-2$, $1-3$, $2-3$ and $3-4$. The breadth-first distance from vertex 0 to vertex 4 is _____.

Q.18 [MSQ • 2 Marks • Moderate]

A connected undirected graph on n vertices is a tree if and only if it

(A) is acyclic and has exactly $n-1$ edges

(B) is connected and has exactly $n-1$ edges

(C) contains at least one cycle

(D) is connected and acyclic

Q.19 [NAT • 2 Marks • Moderate]

Kruskal's algorithm is run on a connected graph with 6 vertices. The number of union operations performed on the disjoint-set structure is _____.

Q.20 [MCQ • 2 Marks • Hard]

A depth-first search of a connected **undirected** graph classifies every edge as

(A) either a tree edge or a cross edge

(B) always a tree edge

(C) either a tree edge or a back edge

(D) a tree edge, a back edge, a forward edge or a cross edge

Q.21 [NAT • 2 Marks • Hard]

The number of edges in the complete bipartite graph $K_{3,4}$ is _____.

Q.22 [NAT • 1 Mark • Moderate]

The maximum number of edges in a simple undirected graph on 6 vertices is _____.

Q.23 [MCQ • 2 Marks • Moderate]

To compute shortest paths from a single source in a graph with non-negative edge weights, the appropriate algorithm is

(A) depth-first search

(B) Kruskal's algorithm

(C) breadth-first search

(D) Dijkstra's algorithm

Q.24 [NAT • 2 Marks • Hard]

The maximum number of edges in a directed acyclic graph on 5 vertices is _____.

Q.25 [MCQ • 2 Marks • Moderate]

The Floyd–Warshall algorithm on a graph with V vertices runs in

(A) $\Theta(V^2)$ time

(B) $\Theta(V^3)$ time

(C) $\Theta((V+E)\log V)$ time

(D) $\Theta(VE)$ time

Q.26 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about graph representations is/are correct?

(A) An adjacency matrix answers "is there an edge $u-v$?" in $O(1)$ time

(B) An adjacency list uses $\Theta(V+E)$ space

(C) An adjacency list answers "is there an edge $u-v$?" in $O(1)$ time

(D) An adjacency matrix uses $\Theta(V^2)$ space

10. Divide and Conquer & Algorithm Design

14 questions • 6 MCQ, 3 MSQ, 5 NAT • Suggested time: 34 minutes

Q.1 [MCQ • 1 Mark • Easy]

The three phases of a divide-and-conquer algorithm are

(A) select, compare, swap

(B) initialise, iterate, terminate

(C) divide, conquer, combine

(D) sort, search, merge

Q.2 [MCQ • 1 Mark • Easy]

Which one of the following is a divide-and-conquer algorithm?

(A) Bubble sort (B) Merge sort (C) Selection sort (D) Counting sort

---

Q.3 [NAT • 1 Mark • Moderate]

A divide-and-conquer algorithm splits a problem of size 16 in half repeatedly until subproblems of size 1 remain. The number of size-1 subproblems is _____.

Q.4 [MSQ • 2 Marks • Moderate]

Which of the following is/are divide-and-conquer algorithms?

(A) Insertion sort (B) Binary search (C) Merge sort (D) Quick sort

---

Q.5 [MCQ • 1 Mark • Moderate]

In merge sort, the "combine" step is

(A) choosing a pivot

(B) splitting the array into halves

(C) merging two sorted halves into one sorted array

(D) swapping adjacent elements

Q.6 [NAT • 2 Marks • Hard]

The maximum-subarray problem solved by divide and conquer satisfies $T(n) = 2T(n/2) + \Theta(n)$. For $n = 8$, the depth of the recursion, counting the top-level call as level 0, is _____.

Q.7 [MCQ • 1 Mark • Moderate]

A greedy algorithm builds a solution by

(A) making the locally best choice at each step

(B) exploring all possible solutions

(C) solving overlapping subproblems and storing the results

(D) randomly sampling the solution space

Q.8 [MSQ • 2 Marks • Moderate]

For which of the following problems does a greedy strategy give a provably optimal solution?

(A) The 0/1 knapsack problem

(B) Minimum spanning tree via Kruskal's algorithm

(C) Single-source shortest paths with non-negative weights via Dijkstra's algorithm

(D) The fractional knapsack problem

Q.9 [NAT • 2 Marks • Hard]

Strassen's algorithm multiplies two $n \times n$ matrices using the recurrence $T(n) = aT(n/2) + \Theta(n^2)$. The value of a is _____.

Q.10 [MCQ • 2 Marks • Moderate]

Binary search is a divide-and-conquer algorithm in which

(A) the combine step dominates the running time

(B) only one of the two subproblems is solved, and the combine step is trivial

(C) both subproblems are solved and then merged

(D) the problem is not divided at all

Q.11 [NAT • 2 Marks • Moderate]

In merge sort on an array of size n, the number of subproblems at recursion depth 4 is _____.

Q.12 [MCQ • 2 Marks • Moderate]

Dynamic programming is preferred over plain divide and conquer when

(A) the problem cannot be divided

(B) the subproblems overlap, so results can be reused

(C) the input is already sorted

(D) the subproblems are completely independent

Q.13 [NAT • 2 Marks • Hard]

For the recurrence $T(n) = 8T(n/2) + n^2$, the solution is $\Theta(n^k)$. The value of k is _____.

Q.14 [MSQ • 2 Marks • Hard]

Which of the following statement(s) about divide and conquer is/are correct?

(A) The master theorem often yields its running time directly

(B) It is naturally expressed using recursion

(C) Its subproblems are independent of one another

(D) It is always more efficient than an iterative solution

Part B — Previous-year questions (with official keys)

These are **genuine questions from the official GATE DA papers** supplied from your archive. Each carries the **official GATE answer key** and its options are kept in the original paper order (not shuffled). Every question is tagged with its exact year and number so you can trust the source. Grouped under the same topics as Part A.

1. Python Language Semantics

Q.1 PYQ

[MCQ • 1 Mark • Moderate]

Consider the given Python program.

```
def append_to_lst(val, lst=[]):
lst.append(val)
return lst

print(append_to_lst(1))
print(append_to_lst(2))
print(append_to_lst(3, []))
```

Which one of the following is the correct output of this program?

(A) `[1]  [2]  [3]`

(B) `[1]  [1, 2]  [3]`

(C) `[1]  [2]  [1, 2, 3]`

(D) `[1]  [1, 2]  [1, 3]`

Source: GATE DA 2026, Q.16 (1 mark) — official paper & official answer key

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Q.2 PYQ

[MSQ • 2 Marks • Hard]

Consider the given Python program.

```
def outer():
x = []
def inner(val):
x.append(val)
return x
return inner

f1 = outer()
f2 = outer()
print(f1(10))  # Line P
print(f1(20))  # Line Q
print(f2(30))  # Line R
print(f1(40))  # Line S
```

Which of the following statement(s) is/are correct?

(A) `f1` and `f2` share the same list `x`

(B) The output of Line Q is `[10, 20]`

(C) The output of Line R is `[10, 20, 30]`

(D) The output of Line S is `[10, 20, 40]`

Source: GATE DA 2026, Q.50 (2 marks) — official paper & official answer key

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3. Recursion & Code Tracing

Q.1 PYQ [MCQ • 2 Marks • Hard]

A recursive function in Python is given.

```
def mystery(n):
if n <= 0:
return 1
else:
return mystery(n-1) + mystery(n-2)

mystery(4)
```

Assume a typical runtime stack is used to manage function calls, each call being pushed onto the stack and removed only after it finishes. The total number of function calls (stack activations), including the initial call, needed to compute `mystery(4)` is

(A) 5 (B) 9 (C) 15 (D) 17

Source: GATE DA 2026, Q.39 (2 marks) — official paper & official answer key

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Q.2 PYQ [MCQ • 2 Marks • Moderate]

Consider the following Python code.

```
def count(child_dict, i):
if i not in child_dict.keys():
return 1
ans = 1
for j in child_dict[i]:
ans += count(child_dict, j)
return ans

child_dict = dict()
child_dict[0] = [1,2]
child_dict[1] = [3,4,5]
child_dict[2] = [6,7,8]
print(count(child_dict,0))
```

Which one of the following is the output of this code?

(A) 6 (B) 1 (C) 8 (D) 9

Source: GATE DA 2024, Q.38 (2 marks) — official paper & official answer key

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Q.3 PYQ [MCQ • 2 Marks • Moderate]

Consider the following Python function.

```
def fun(D, s1, s2):
if s1 < s2:
D[s1], D[s2] = D[s2], D[s1]
fun(D, s1+1, s2-1)
```

What does this function `fun()` do?

(A) It finds the smallest element in `D` from index `s1` to `s2`, both inclusive.

(B) It performs a merge sort in place on the list `D` between indices `s1` and `s2`, both inclusive.

(C) It reverses the list `D` between indices `s1` and `s2`, both inclusive.

(D) It swaps the elements of `D` at indices `s1` and `s2` and leaves the rest unchanged.

Source: GATE DA 2024, Q.41 (2 marks) — official paper & official answer key

4. Asymptotic Complexity & Recurrences

Q.1 PYQ

[MCQ • 1 Mark • Hard]

Quick sort is used to sort an array of n distinct, randomly ordered elements. In every call the pivot is chosen as the *first* element of the current subarray. Let $T(n)$ denote the expected time to sort the array, and assume partitioning takes time linear in the size of the current subarray. Which recurrence correctly represents $T(n)$?

(A) $T(n) = T(1) + T(n-1) + O(n)$

(B) $T(n) = T\left(\frac{n}{4}\right) + T\left(\frac{3n}{4}\right) + O(n)$

(C) $T(n) = 2T\left(\frac{n}{2}\right) + O(n)$

(D) $T(n) = \frac{1}{n}\sum_{k=0}^{n-1}[T(k) + T(n-k-1)] + O(n)$

Source: GATE DA 2026, Q.15 (1 mark) — official paper & official answer key

5. Searching & Sorting

Q.1 PYQ

[NAT • 1 Mark • Moderate]

Let A be a sorted array containing 1000 distinct integers. A recursive binary search is performed on A to find an element y. Each comparison checks whether the middle element computed in the current step is equal to, less than, or greater than y. The maximum number of comparisons that may be performed if y is **not** an element of A is _____.

Source: GATE DA 2026, Q.31 (1 mark) — official paper & official answer key

Q.2 PYQ

[NAT • 2 Marks • Hard]

Consider the given Python program.

```
def fun(L, i=0):
if i >= len(L)-1:
return 0
if L[i] > L[i+1]:
L[i+1], L[i] = L[i], L[i+1]
return 1+fun(L, i+1)
else:
return fun(L, i+1)

data = [5, 3, 4, 1, 2]
count = 0
for _ in range(len(data)):
count += fun(data)
print(count)
```

The output of the program is _____.

Source: GATE DA 2026, Q.58 (2 marks) — official paper & official answer key

Q.3 PYQ [MCQ • 2 Marks • Hard]

Consider the following sorting algorithms: **(i)** Bubble sort, **(ii)** Insertion sort, **(iii)** Selection sort. Which choice of algorithms sorts the array $[4, 3, 2, 1, 5]$ into increasing order after **exactly two passes** over the array?

(A) (i) only (B) (iii) only (C) (i) and (iii) only (D) (ii) and (iii) only

Source: GATE DA 2024, Q.45 (2 marks) — official paper & official answer key

Answer Key

Answers only. A key with several letters (e.g. **BD**) is an MSQ, where every listed option must be selected and no other. For NAT questions any value within the usual GATE tolerance of the figure shown is accepted.

Part A — Practice

1. Python Language Semantics

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	4	**2**	A	**3**	2	**4**	A	**5**	2
6	D	**7**	7	**8**	ABD	**9**	3	**10**	D
11	4	**12**	D	**13**	3	**14**	ACD	**15**	1
16	A	**17**	2	**18**	4	**19**	C	**20**	3
21	D	**22**	5	**23**	ABD	**24**	3		

2. Python Built-in Data Structures

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	3	**2**	B	**3**	2	**4**	A	**5**	6
6	A	**7**	3	**8**	ABD	**9**	4	**10**	A
11	5	**12**	4	**13**	D	**14**	3	**15**	10
16	B	**17**	2	**18**	ABC	**19**	1	**20**	3
21	A	**22**	9	**23**	C	**24**	4		

3. Recursion & Code Tracing

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	120	**2**	B	**3**	8	**4**	15	**5**	6
6	D	**7**	29	**8**	3	**9**	31	**10**	B
11	15	**12**	A	**13**	ABD	**14**	15	**15**	4
16	A	**17**	1024	**18**	8	**19**	B	**20**	55
21	BCD	**22**	3						

4. Asymptotic Complexity & Recurrences

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	**2**	6	**3**	B	**4**	32	**5**	A
6	D	**7**	15	**8**	AC	**9**	55	**10**	B
11	108	**12**	A	**13**	10	**14**	ABC	**15**	10
16	B	**17**	24	**18**	B	**19**	31	**20**	D
21	385	**22**	ABC	**23**	16	**24**	C		

5. Searching & Sorting

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	4	**3**	D	**4**	10	**5**	ABC
6	15	**7**	C	**8**	3	**9**	6	**10**	C
11	10	**12**	ACD	**13**	100	**14**	B	**15**	3
16	C	**17**	3	**18**	2	**19**	D	**20**	148
21	ABC	**22**	6	**23**	A	**24**	4	**25**	D
26	3	**27**	A	**28**	ABD	**29**	5	**30**	A

6. Stacks, Queues & Linked Lists

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	C	**3**	1	**4**	7	**5**	D
6	3	**7**	ABC	**8**	A	**9**	16	**10**	B
11	2	**12**	D	**13**	2	**14**	7	**15**	BCD
16	4	**17**	3	**18**	C	**19**	10	**20**	B
21	100	**22**	0	**23**	ACD	**24**	24		

7. Trees & Binary Search Trees

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	**2**	15	**3**	C	**4**	4	**5**	2
6	D	**7**	BC	**8**	3	**9**	C	**10**	14
11	42	**12**	A	**13**	9	**14**	3	**15**	ABC
16	16	**17**	C	**18**	19	**19**	6	**20**	A
21	6	**22**	3	**23**	BCD	**24**	40		

8. Hashing

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	**2**	3	**3**	B	**4**	5	**5**	A
6	0.75	**7**	ABD	**8**	4	**9**	B	**10**	2
11	4	**12**	D	**13**	3	**14**	BCD	**15**	2
16	D	**17**	3	**18**	B	**19**	2	**20**	ACD

9. Graphs & Graph Algorithms

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	28	**3**	C	**4**	D	**5**	11
6	8	**7**	ABC	**8**	10	**9**	A	**10**	8
11	10	**12**	A	**13**	2	**14**	ACD	**15**	10
16	C	**17**	3	**18**	ABD	**19**	5	**20**	C
21	12	**22**	15	**23**	D	**24**	10	**25**	B
26	ABD								

10. Divide and Conquer & Algorithm Design

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	B	**3**	16	**4**	BCD	**5**	C
6	3	**7**	A	**8**	BCD	**9**	7	**10**	B
11	16	**12**	B	**13**	3	**14**	ABC		

Part B — Previous-year questions (official keys)

Topic	Q	Official answer
GATE DA 2026, Q.16 (1 mark)	T1.1	**B**
GATE DA 2026, Q.50 (2 marks)	T1.2	**BD**
GATE DA 2026, Q.39 (2 marks)	T3.1	**C**
GATE DA 2024, Q.38 (2 marks)	T3.2	**D**
GATE DA 2024, Q.41 (2 marks)	T3.3	**C**
GATE DA 2026, Q.15 (1 mark)	T4.1	**D**
GATE DA 2026, Q.31 (1 mark)	T5.1	**10**
GATE DA 2026, Q.58 (2 marks)	T5.2	**8**
GATE DA 2024, Q.45 (2 marks)	T5.3	**B**

Score sheet

Anything below 70% on a topic means the concept, not the practice, is the problem — go back and revise before re-attempting.

Topic	Questions	Correct	Time taken
Python Language Semantics	24		
Python Built-in Data Structures	24		
Recursion & Code Tracing	22		
Asymptotic Complexity & Recurrences	24		
Searching & Sorting	30		
Stacks, Queues & Linked Lists	24		
Trees & Binary Search Trees	24		
Hashing	20		
Graphs & Graph Algorithms	26		
Divide and Conquer & Algorithm Design	14		
Total	**232**		